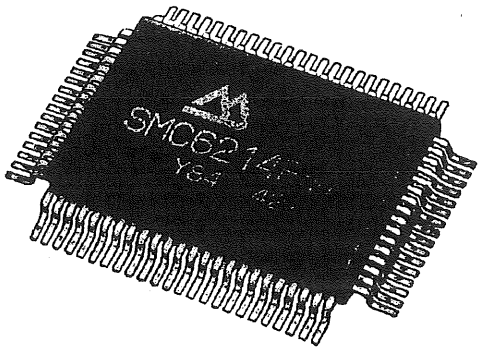
INTRODUCTION TO SIGNALLING

10

SOLID STATE INTERLOCKING



A.J.R.Rowbotham

INTRODUCTION

Over the years lever frames with mechanical locking have been gradually replaced by relay interlockings operated from control panels. This situation started to change again in the late 1980s when a solid state interlocking system was introduced. This system was initially developed by BR Research at Derby, starting in the late 1970s. It was realised as the development went on that BR Research did not have the resources or the skills to turn their theoretical ideas into a practical, usable and manufacturable system. A tripartite agreement was drawn up between BR, the then GEC General Signal Co. and the then Westinghouse Brake and Signal Co. to produce such a workable system. The prototype SSI system was installed and commissioned, again as a joint exercise, at Leamington Spa in September 1985.

The name *Solid State Interlocking (SSI)* is used for this BR system, but unhelpfully this has also become a generic term for any solid state system in the same way as the words 'Hoover' and 'Xerox' have become generic. There are arguments that it should have been called 'Solid State Signalling System' as the BR SSI embraces the trackside interfaces as well as the interlocking itself. Any mention of SSI refers to the BR design unless otherwise stated.

THE INTERLOCKING

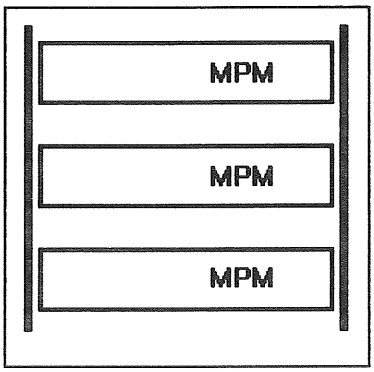
The SSI equipment for one interlocking consists of several modules housed in a standard '19 inch' cubicle.

The Multiprocessor Module

The module that carries out the interlocking function is termed the *Multiprocessor Module (MPM)* and three are required in the one cubicle. Each module has identical hardware and software, and each has four microprocessors incorporated.

One processor in each module processes the interlocking logic. It is also involved with communicating with its equivalent in the other two MPMs. This liaison is how the safety of the SSI is ensured. Each MPM communicates continuously with its two partners, comparing their workings and memory contents. The normal way of working is for the three to work together and to be making the same decisions from the same information.

If however, one of the MPMs detects that it is behaving differently from its two partners then it will "commit suicide" by deliberately blowing its own internal security fuse to take itself out of circuit (and bring up an alarm). At the same time, the two MPMs working together will "assassinate" the odd one out by blowing its fuse for it. In either case the system continues working on the other two, and as long as they agree there is no problem. However, if these remaining two disagree then the whole system must close itself down as there is no way of telling which is in error. This type of system is known as a *two-out-of-three voting system.*



Once one MPM has a blown fuse it is important that the module is replaced rapidly before the two remaining working MPMs can disagree and shut the system down. The system is especially designed so that the faulty MPM can be quickly unplugged and a new one plugged in without affecting the working of the remaining two. The replaced MPM gets itself in step automatically with its two partners and the system settles down again with the three MPMs working together.

The second and third microprocessor in each MPM organises the communications with the trackside equipment, described later.

The fourth microprocessor in each MPM organises the communications with other SSI systems, described later.

Software

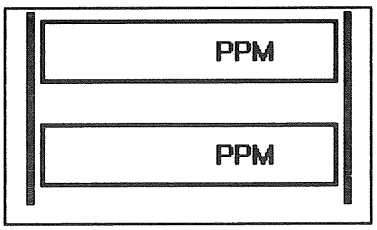
Notwithstanding the triplicated MPMs, the safety of the system depends on the safety of the software that works it. Software for safety systems has to go through a formal process known as *Validation*, to obtain assurance that it will behave in a safe manner when run. This validation for the SSI software involved well over 10,000 pages of documentation and many man-years of effort. No two signalling schemes are the same, and to carry out this validation exercise for every interlocking would render the use of SSI uneconomic. To overcome the problem the software is divided into two parts - the *fixed program* and the *data.*

The fixed program is common to all systems and thus only needs to be validated once. Railtrack are responsible for this program and distribute it to the contractors.

The Data is the portion of the software that engineers the common program to work for a particular section of railway. The data is (usually) produced by the contractors and consists of many hundreds or even thousands of statements defining the interlocking controls required. These statements are written in a 'high level' language, that is one specially devised so that Signal Engineers readily understand it, using signalling terms, without having to be programmers or particularly computer literate.

Panel Processor Modules

The *Panel Processor Module (PPM)* acts as the interface between the signaller and the MPMs. It is not a safety component of the SSI, so does not need to be triplicated like the MPM, but is duplicated, solely to improve availability.



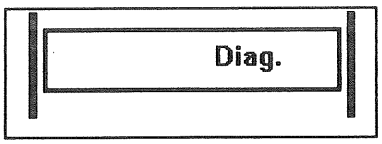
The PPM is designed to work directly via S*erial links* with an Integrated Electronic Control Centre system (IECC), or with a 'hard panel' containing individual buttons, switches and lamps.

If connected to an IECC system, both PPMs work in parallel. The IECC subsystems each select a PPM to act as a 'master', but also maintain regular contact with the other PPM, which can then become the master in the event of a failure.

If connected to a hard panel, then duplicated panel multiplexers (PMUX) are required at the panel end. These act as a serial/parallel interface, to convert the serial stream of data in the link into the individual parallel connections to and from the lamps and buttons/switches. In this case both PPMs are working in parallel with the indications passing to both PMUXs and the PMUXs actually decide which PPM they work with, formally changing over if one PPM fails. The controls are passed permanently in parallel from the PMUXs to the two PPMs.

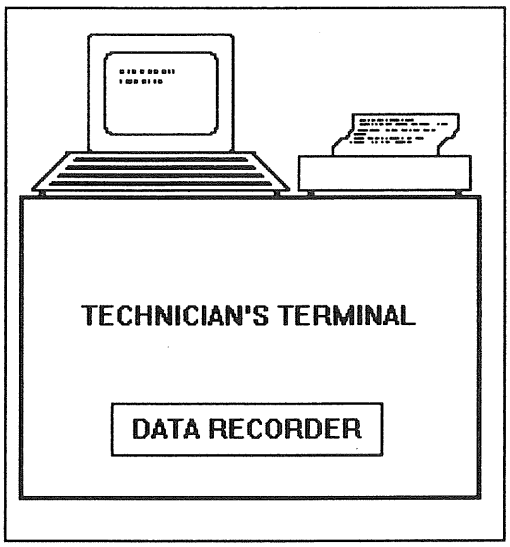
Technician's Terminal and Diagnostic MPM

If there is a problem in a relay interlocking fault finding is fairly easy. Relays can be seen to be energised or de-energised and voltages can be measured on every relay connection. With a microprocessor interlocking, however, everything is hidden away inside semiconductor chips making the equivalent sort of fault finding impossible.



The *Technician's Terminal (TT)* is a separate desk including a VDU, Keyboard and printer that enables the technician to effectively monitor the SSI workings. A third type of module, the *Diagnostic Module*, monitors all that is going on in the system and passes information to the TT. The Diagnostic Module hardware is identical to the MPM but has different software.

All faults are instantly printed out on the printer, and may also be displayed on the TT VDU if required. These faults include not only SSI system faults, but signalling faults such as filament and lamp failures in signals, detection failure from points etc. The Technician is able to impose extra controls on the interlocking, e.g. holding a signal at danger, locking a set of points in one direction. These additional controls , akin to *collaring* on a conventional control panel, could be used when engineering work is taking place on the line, for instance, to protect staff. What the Technician cannot do is to override any of the normal interlocking features - that would be very dangerous indeed and quite unacceptable.

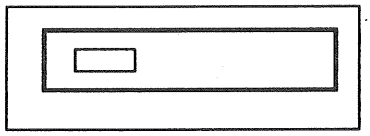


Also incorporated in the TT is the 'Black Box' - a tape *Data Recorder* unit that records everything that happens in the system with a time stamp added. There is thus a complete record on tape of every control the signaller enters, records of track circuit occupation and clearing, signal aspects changing, points moving etc. If there is an incident the tape contents can be analysed and the exact sequence of events determined. Drivers can no longer pass a red signal and then try to claim the signal was green. The tape data would clearly show the claim to be false. Each 'black box' has two tape cartridges, each with at least 24 hours capacity. The tapes change over automatically at midnight each night. If, however, the density of traffic was such that the tape was full before the midnight changeover, then the change would be made instantly so no data was lost. The tapes are used continuously, simply overwriting what is already on the tape. If an incident happens, the tape needs to be retrieved fairly quickly so there is no chance of the data being overwritten.

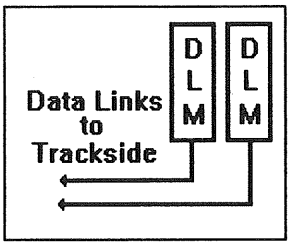
Memory Modules

The program and data for a working system are stored in semiconductor memory chips - EPROMs. On the first design of modules the EPROMs were plugged directly into sockets on the printed circuit boards and the module was then sealed. This meant that if a change was needed, the MkI modules had to be sent back to the manufacturer, the seals broken, the EPROMs changed and the unit then resealed.

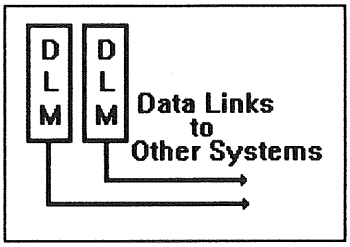
On the current (MkII and upwards) design, the EPROMs are housed in a *Memory Module* that slides into the front of a module like a little drawer. All the necessary electrical connections are made via a detachable flexible ribbon cable.



Data Link Modules



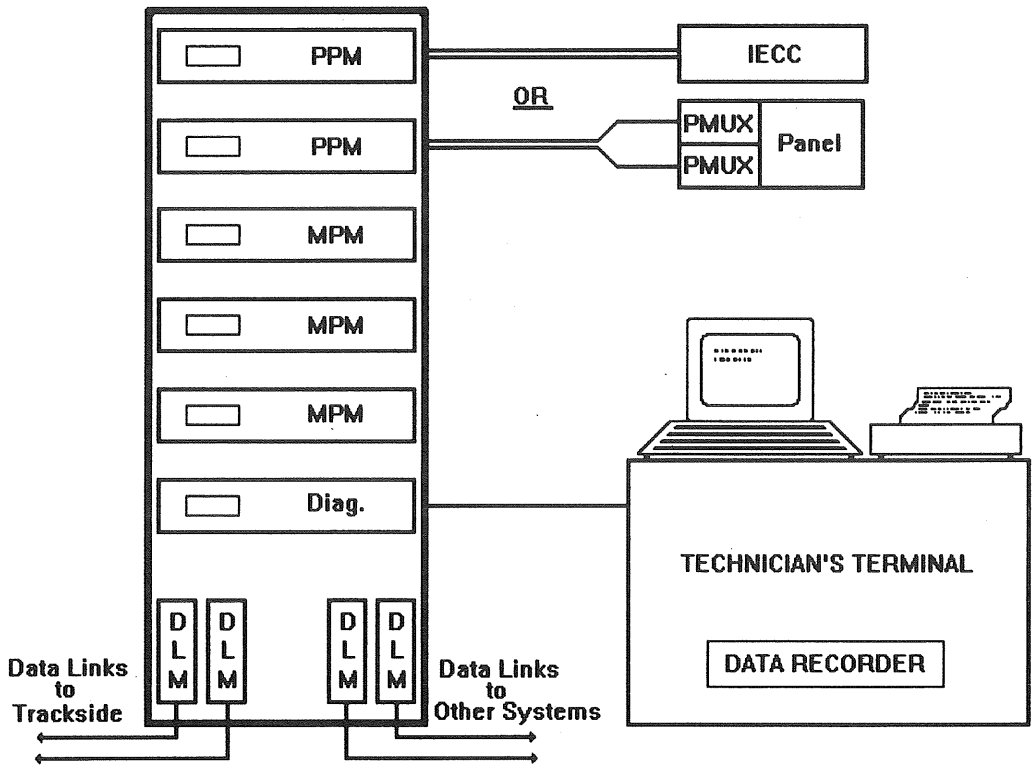
The *Data Link Module (DLM)* acts as an interface between the interlocking and the *Data Link Cables* that communicate with the trackside equipment over the *Trackside Data Link*. Two such 2 core twisted pair cables are provided. Like the PPMs two are provided for reliability purposes, but each is considered a 'safe item' in its own right. The DLMs are purely hardware devices requiring no software - program or data.



On anything but the smallest schemes more than one SSI interlocking cubicle will be necessary to cover the area that is required to be controlled. These various interlockings will need to exchange information between them relating to the area of the boundaries between them. This communication path, the *Internal Data Link*, is again by means of duplicated data links, and a pair of 'star points' are provided that all such links are connected to, up to a maximum of 30 interlockings. Connections to the star points are also made through DLMs, thus most interlocking cubicles would house 4 DLMs - two for internal and two for external communications.

The Interlocking Equipment - Summary

The complete interlocking cubicle, and Technician's Terminal assembly is shown below:



TRACKSIDE EQUIPMENT

The modules that interface with the trackside signalling equipment are normally housed in conventional location cases and are known as *Trackside Functional Modules (TFMs).* Two types are available, the *Signal Module* (strictly speaking a general module) and a *Points Module* (specially designed for one purpose only).

Each TFM in an SSI system regularly receives a message incorporating controls from the interlocking, and sends indications back in a return message. A maximum of 63 TFMs are allowed in one system, with identities numbered from '1' to '63'. This is the principal parameter in deciding how many SSI systems might be needed for a particular scheme. In practice the full number of 63 would not be used - there must be spare capacity allowed, frequently 10%. Signal and points TFMs may be mixed in any combination.

In the full cycle the interlocking sends a message (or *telegram*) to TFM 1, receives a message back, sends to TFM 2, receives a message back etc., until all 63 have been contacted. The cycle then starts again at TFM 1. This complete cycle takes about ¾ of a second. Messages are thus passing alternately in either direction in the data link cable - a *half-duplex* system.

The hardware of the TFMs is duplicated within the same module. The two parts compare their workings continuously and if any discrepancy is noticed the TFM is taken out of service by blowing its own internal *security fuse*. The state of the TFM at the time of failure is stored in non-volatile memory, which can then be interrogated when the module is returned to the factory. Notice that this two-out-of-two voting system is less rigorous than the two-out-of-three voting system used on the three separate MPMs in the interlocking. The argument being for this that if the MPMs fail completely, then the whole interlocking is lost, if the TFM fails then only one or two functions are lost. It was not considered worth while to go to the expense of internally triplicating the TFMs.

It is important to be able to change a failed TFM very quickly to regain the lost functions that it controls. There are no data or program EPROMs to worry about, so that makes for easy replacement. However there does remain the identity number of the TFM. A common practice for this would be to have a set of thumb wheel switches, or perhaps DIP switches to set to represent the number. However this identity setting would have to be remembered when a TFM was replaced. If it was forgotten then chaos would ensue, possibly leaving two TFMs with the same number. All the wiring to the TFM is through one multi-way plug, with the matching socket on the module. Incorporated into the plug wiring are a set of pins looped together by wires in a particular way to represent the identity number of that module. Thus to change a module, the plug is disengaged from the socket, a new TFM mounted on the rack, and the plug (incorporating the identity) plugged back in. There is no need to worry about setting any switches of any sort.

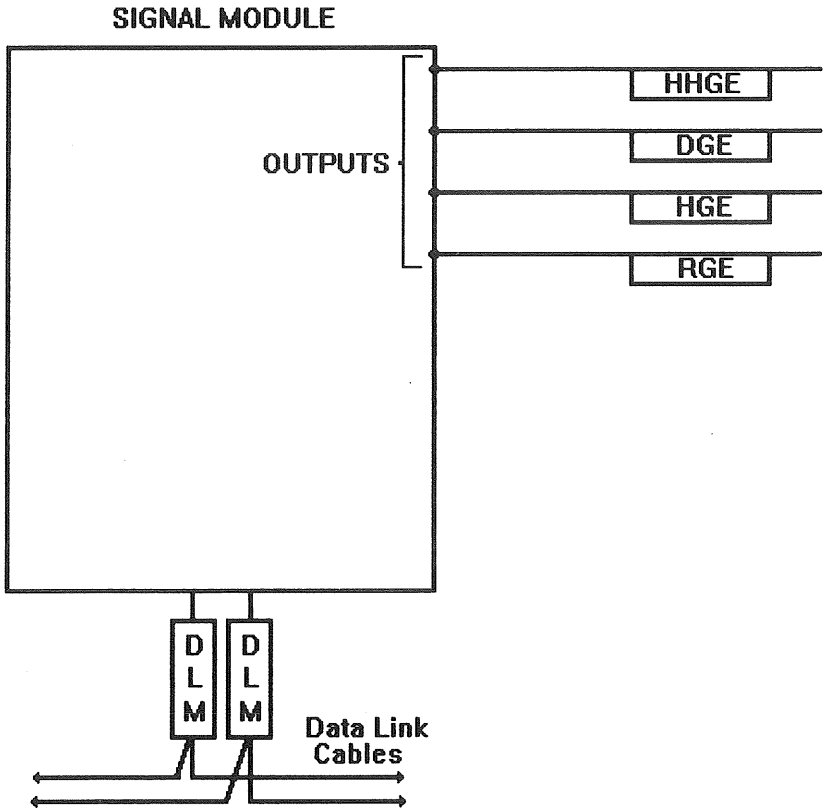
The Signal Module

The signal module has eight a.c. outputs, each of which may be switched on or off under instruction from the interlocking. The telegram received once each cycle from the interlocking consists of a command for the eight outputs, instructing each to be switched on (energised) or switched off (de-energised). The outputs may be used to directly drive signal lamps (i.e. a three aspect signal would need three outputs), AWS magnets etc. A specially designed BR 930 style safety relay is available for energising from an output, if required for uncommon applications.

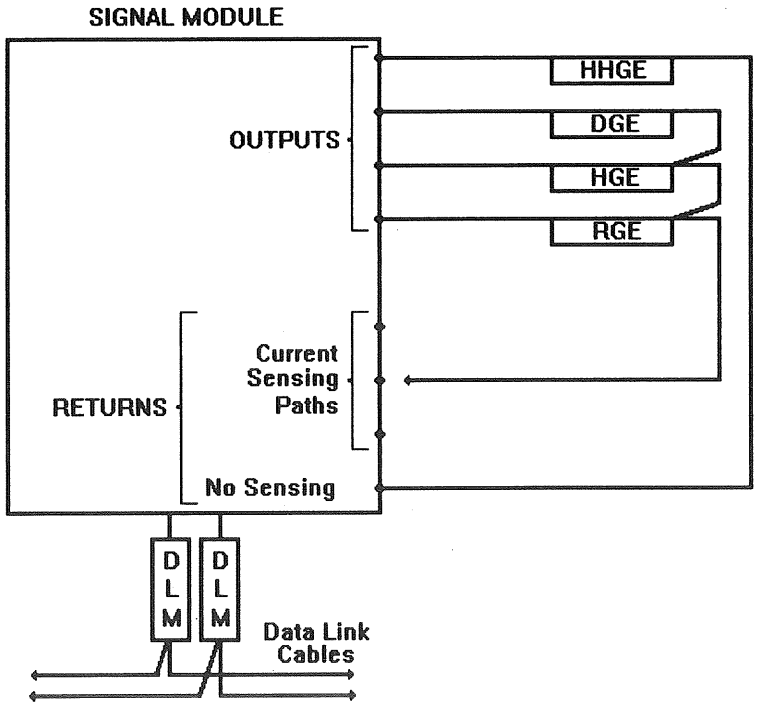
An important safety feature is that it is possible to sense if current is flowing from an output. If a current flow is detected, a check is made by the module that that output has in fact been instructed by the interlocking to be switched on. If it has not, then the module goes into the temporary *output disabled* mode for 8-10 seconds. In this state telegrams are still received and sent. If the problem does not resolve itself the outputs are permanently disabled and an appropriate alarm returned to the interlocking in the next telegram.

The TFMs are connected to the two data link cables through DLMs identical to those in the interlocking cubicle . Up to five TFMs in one location area may be interfaced through one set of DLMs. Each DLM has five sets of terminals provided for these TFM connections.

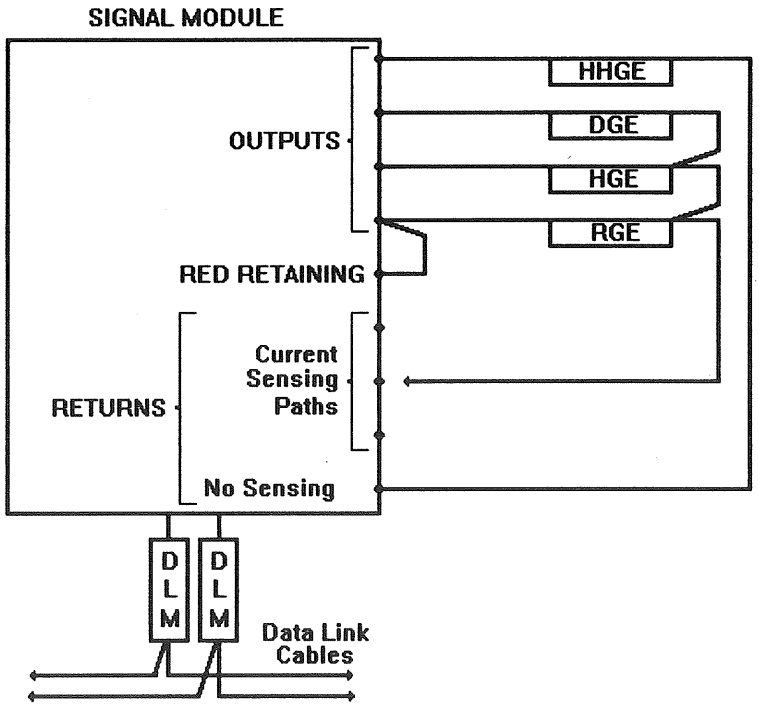
Considering a four aspect signal, four outputs out of the eight available have to be selected to drive the four signal lamps.



If the signal was relay driven a lamp proving relay would be wired in to prove that either the red, first yellow or green lamp was lit. The equivalent function incorporated in the signal TFM is termed *current sensing*, and three different types of current sensing paths are provided. One is designed to prove that a main signal lamp is lit, one to prove the five lamps of a junction direction indicator, and the third to prove the lamps of a stencil indicator. Each is bought out to a terminal so the return from the signal must be connected to the correct terminal for the appropriate sensing path. The return from the second yellow has to be brought back to the return terminal without any internal current sensing path.

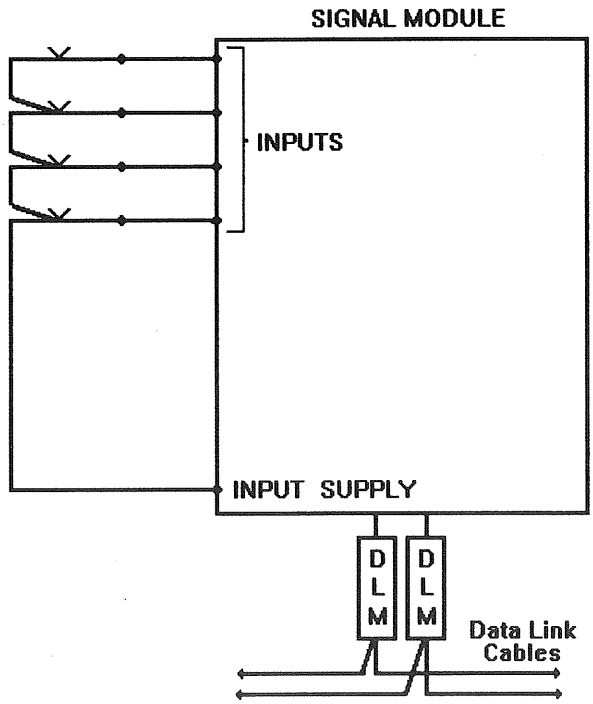


If the TFM were to blow its fuse then unless special arrangements were made, the signal would be dark as no outputs would be energised. In fact there is a special feature, *red retaining*, provided where a terminal (not one of the eight outputs) is energised especially, when the security fuse blows. This red retaining terminal is connected by a wire loop to the conventional output terminal driving the most restrictive aspect of the signal (usually red). Thus, if the TFM takes itself out of service the red aspect is 'false fed' from the red retaining terminal.



As one Signal Module could control two simple signals, there are in fact two independent red retaining outputs and sufficient current sensing paths for either two simple main signals or one signal with indicators.

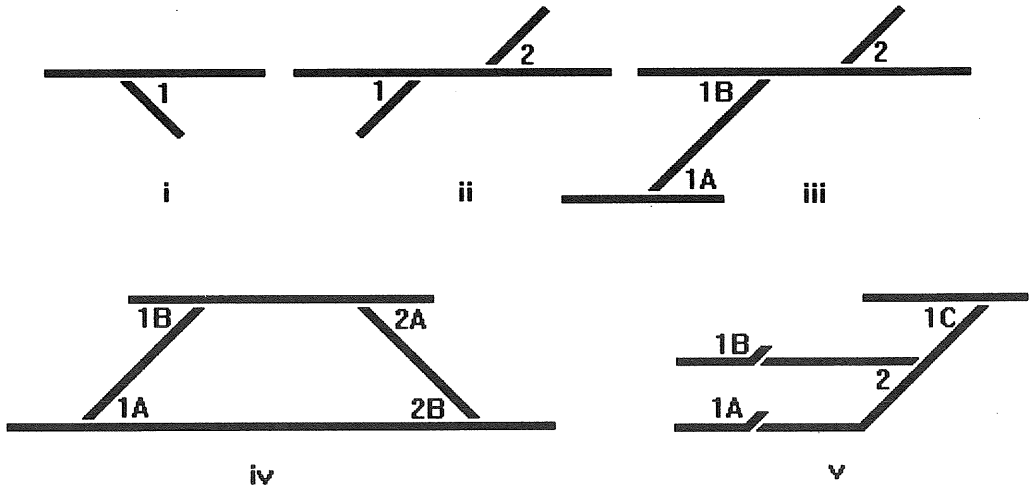
As well as receiving instructions for the eight controls from the interlocking, the module also has the facility for returning eight indications back to the interlocking. Only six of these are available for general use as two are reserved for return indications relating to the current sensing paths. These remaining six can be used as required for any other indication, the most common being the return of track circuit information from track relays housed in the same location. An input is fed over the relevant relay contact into the appropriate input circuit.



The supply for these input circuits is provided from the module and incorporates a special safety feature. The input circuit supply is frequency modulated in a pseudo random sequence and the inputs are checked for identical modulation. This ensures that the input circuits are definitely being fed from the actual module, and have not picked up a spurious voltage from some other source. Two supplies are provided - an internal one for feeding over contacts of relays in the same case, and an external one for circuits feeding outside the case and back, such as collecting filament failure information from a signal head.

The Points Module

The Points Module, as the name implies, is specially designed to work points. Basically there are eight outputs and eight inputs. The eight outputs are designed to work the pumps and valves associated with clamp lock drives. A maximum of up to four point ends split between two identities can be driven from one module - either two double ended sets with different identities or a combination of three machines with one identity and a single machine on the other. Any of the following groupings of points could therefore be worked from one Points Module:

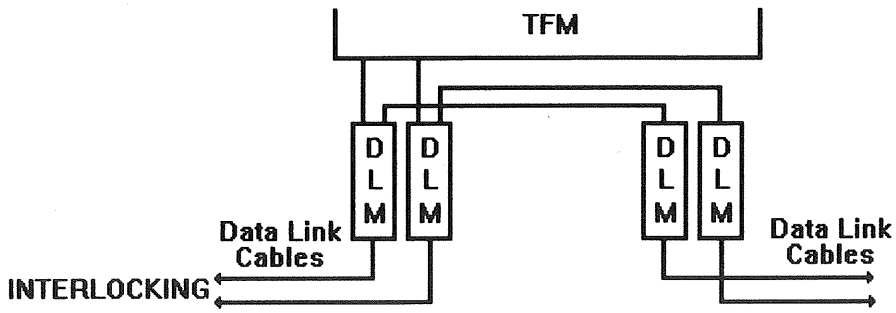


Four of the inputs are internally connected for the normal and reverse detection of the two point identities. This leaves four inputs for general use.

If electric point machines are to be worked from an SSI then interface safety relays must be used between the Points Module and the machines. The direct outputs from the Points Module have voltages and current capacities designed specifically for Clamp Lock control.

THE DATA LINK

Because of the characteristics of the cable used in the Data Link the length of a simple system is limited to 10 km. This is rather restrictive, but if *repeaters* are used, this length may be increased to a more practical maximum. The repeater boosts the level of the telegrams in the link, overcoming losses occurring in the cable, and consists of two DLMs connected back to back. Consequently, no new physical device is required. At a suitably placed TFM an extra DLM is added to each cable to form the back to back arrangement required for the repeater function.



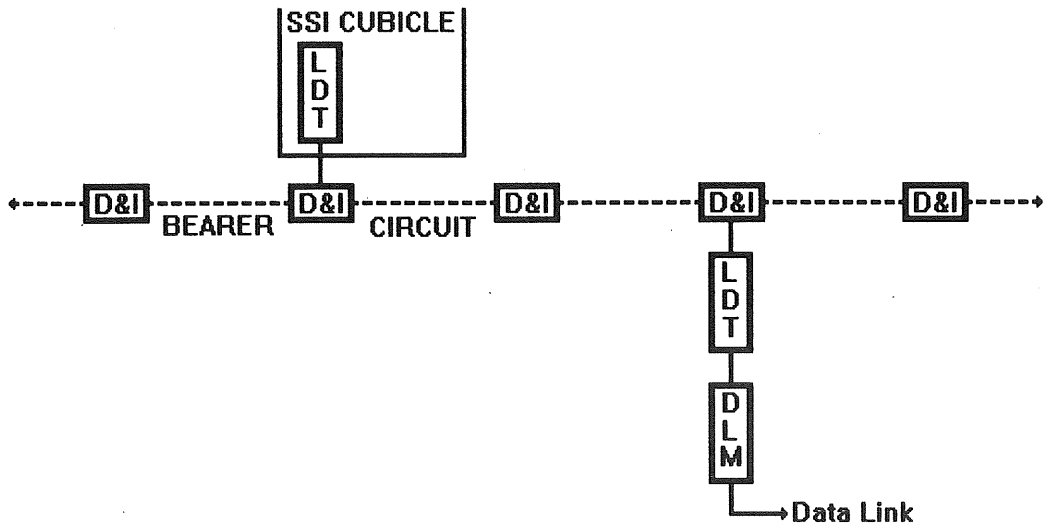
If four repeaters are incorporated at roughly 8 km intervals, then a maximum length of 40 km is allowed.

If a distance of more than 40 km is required then a different technique - *a bearer circuit* - is used. A bearer circuit is a general purpose telecommunications system, usually not provided by the Signalling function, which the Signal Engineer may use for the block transfer of SSI data over long distances. The data may well be multiplexed with other information in the system, but that does not cause any concern.

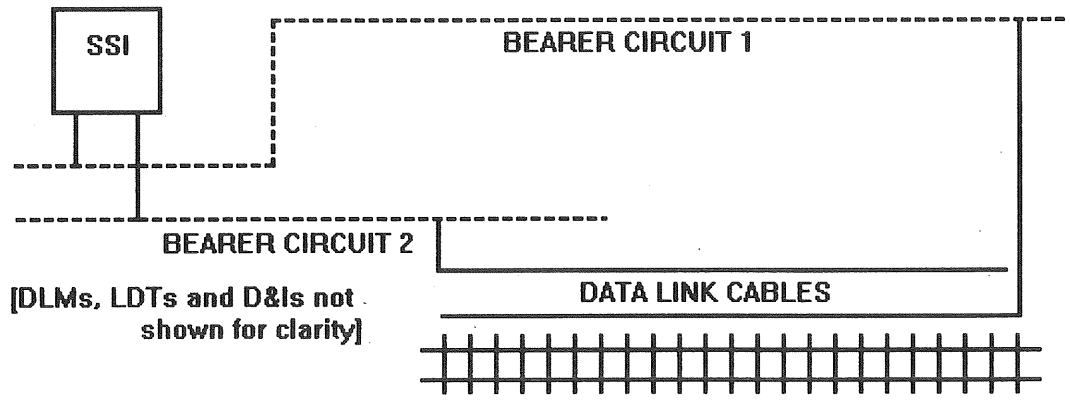
The places where data may be put in to, or taken out of, the system are known as a *Drop & Insert (D&I)* points. These would normally be found in telecoms buildings or huts close to the trackside. However, the physical links between them might well be along a completely diverse path away from the railway infrastructure. The transmission path might be copper or fibre optic or even microwave. Again, it does not concern the Signal Engineer which is used.

The Pulse Code Modulation system used in the bearer link is not directly compatible with the protocol used in the SSI Data Link, so a *Long Distance Terminal (LDT)* is needed to interface the two systems together.

The diagram below shows the connection of one Data Link to the bearer circuit, with an LDT, rather than a DLM, interfacing the interlocking cubicle to the telecomms path at a D&I point. The data is extracted from the system at a suitable D&I point to start the journey along the trackside in the area being controlled. At this end, an LDT interfaces between the system and the DLM. From there on the system reverts back to the normal SSI Data Link cable and protocols.



The duplicate Data Link might be carried in the same bearer circuit, or preferably be taken by a different circuit on a completely diverse path. It might even be taken to the far end of the section of railway concerned, and be fed back in the opposite direction along the track. This would give the maximum availability under various fault conditions.



THE DESIGN WORKSTATION

The SSI data is prepared at a *Design Work Station (DWS)* which consists of a central computer, keyboards, VDUs and a number as required of SSI interlocking Simulation Cubicles. Data is entered into the system on a keyboard and editing facilities are available. The data statements represent the logic defining the complete interlocking requirements for the area concerned. The rules and syntax requirements are laid out in Railtrack documents. A very simple data statement might be:-

P1123N , TGUL c , TGDL c

This statement says that points (P) 1123 cannot be driven from reverse (understood) to Normal (N) unless track circuits (T) GUL and GDL are both clear (c).

When the many thousands of lines of data are complete it is *compiled.* This is the software process of converting the data from the high level language used by the Signal Engineers into the binary coding that the SSI processors require. When successfully compiled and then 'desk' checked (a visual check of the data from a printout) the data is 'blown' into EPROMs for use in the various interlocking modules.

An important feature of the DWS is the Simulation Cubicle which enables a full functional test of the interlocking to be made in the office, before installation on site. Two Colour Displays (VDUs), and two trackerballs are required. One display incorporates all the controls and indications that a signaller would normally have to operate the interlocking. The other display shows the state of all the inputs from, and outputs to, the TFMs and allows tracks to be operated and faults deliberately to be introduced to enable a comprehensive test to be carried out. The Simulation cubicle incorporates one PPM, one MPM, one Diagnostic MPM and one Simulator MPM.

OTHER SYSTEMS

The SSI system described above uses a two-out-of-three voting system to achieve the required level of safety. Manufacturers of solid state interlockings in other countries have used different methods of ensuring safe systems. A selection demonstrating these differing methods are briefly described below.

Ebilock

The Adtranz system, termed 'Ebilock', uses computer units containing three processors. Two of these carry out the signalling tasks, using different programs to carry out the same tasks. Both have to agree for a function to be actuated. If there is a disagreement then an alarm is raised and control is passed to a second computer, working as a hot standby. This second unit is continuously updated with information, so it can take over instantly without affecting the interlocking. The third processor in each computer unit deals with non-vital functions related to servicing and interfaces.

The equivalents to the TFMs are termed *'Object Controllers'*, a non railway-specific term. The Object Controllers may be distributed along the trackside, or concentrated in the signal box with the interlocking.

The data is prepared with the help of a tool, 'Ebitool', which actually has the rules of interlocking built in. Data can then be produced automatically from the track plan which is input into the system with other relevant information.

VPI

One of the more successful competitive systems, developed by the General Railway Signal Co. in the USA works on a completely different principle. Called Vital Processor Interlocking (VPI) this system uses 'Numerically Integrated Safety Assurance Logic' - a complex safety software idea that does not require triplicated processors (although they may be duplicated for availability). The system is designed to directly feed the outside equipment from the interlocking, eliminating the need for TFMs. As the logic operations are processed a 'check word' is built up. As long as this check word is correct the system assumes that all is well. If anything goes wrong, the correct check word is not generated and the system is closed down. In fact a single safety relay is kept energised by the repetitive generation of the correct check word. Contacts on this relay feed plain power to the output boards, so if a problem is detected, the relay will be de-energised and all interlocking outputs are immediately de-energised (so all signals go dark). There is no signalling logic in the program, and the system is designed solely to evaluate Boolean logic expressions.

WESTRACE

Westinghouse Signals together with their associated companies in Spain, Australia and the USA have developed the 'WESTRACE' system. Each interlocking is made up from a collection of printed circuit board modules selected from the collection of:

- Vital Logic Module.
- Vital Parallel Input Module.
- Vital Lamp Output Module.
- Vital Relay Output Module.
- Vital serial Input/output Module.
- Non-vital Communications Module.
- Diagnostics Module.

The Vital Logic Module consists of three boards which together ensure the safety of the system. The system employs diverse assurance for the safe operation of each processor. The integrity of each processor is proved, not only by its own tests, but also by tests performed on it by its neighbours. Each dual proven processor runs two diverse programs for its operation, achieving the safety of a dual system, but with only one set of hardware. In addition to checksum protection, the application logic operates in two parallel channels, one holding 'true' and the other 'complement' forms of the data. Comparison between the channels takes place at each stage of the processing.

Vital input and output Modules have the capacity to isolate individual faulty inputs and outputs so that the remainder of the Module can continue to operate, giving 'graceful degradation'.

If there is a fault that cannot be isolated by graceful degradation, then system shutdown is implemented by two independent methods, which results in the power being removed from all vital outputs.

CONCLUSION

The BR developed SSI has been an extremely successful product with several hundred systems in use world wide. Other organisations have developed systems in competition, but have nowhere near the same number of systems in use.

In general the systems installed overseas are the standard system as used in the UK. However, a number of systems have been installed in Belgium, using a different TFM. system principally to suit the more complex signal requirements of the SNCB/NMBS. The Belgian system interfaces a conventional UK interlocking to these TFMs using a fibre optic data link rather than a copper one. This has the big advantage of eliminating electrical interference in the data link cables that might have a detrimental effect on the working of the system. Each optical data link is a loop, and the two loops are fed in opposite directions. In this way, even if both optic cables are cut at the same spot, all TFMs will continue to receive and send telegrams. The Optic Data Link Modules (ODLMs) receive all the telegrams in the optic cable, pass the relevant telegrams to the connected TFMs, and then regenerate the message to pass to the next ODLM around the loop. There is also a Belgian version of the electrical UK DLM, the EDLM, which is required for the communication between SSI cubicles. An interlocking cubicle would consequently contain to ODLMs for external communication and two EDLMs for internal communication.

The French SNCF has now started installing SSI systems, again using the Belgian style TFMs, ODLMs and EDLMs.